

1 Solution — GIAM 6.6

Problem 7

1.1 Problem

If S and T are subsets of a set D , what is the product of their characteristic functions $1_S \cdot 1_T$?

1.2 Setup

The characteristic function of a subset $A \subseteq D$ is

$$1_A : D \rightarrow \{0, 1\}, \quad 1_A(x) = \begin{cases} 1 & x \in A, \\ 0 & x \notin A. \end{cases}$$

The pointwise product $1_S \cdot 1_T : D \rightarrow \{0, 1\}$ is defined by

$$(1_S \cdot 1_T)(x) = 1_S(x) \cdot 1_T(x).$$

1.3 The Product Is the Indicator of the Intersection

For any $x \in D$, the product $1_S(x) \cdot 1_T(x)$ is 1 iff both factors are 1:

$$1_S(x) \cdot 1_T(x) = 1 \iff (1_S(x) = 1) \text{ and } (1_T(x) = 1) \iff x \in S \text{ and } x \in T \iff x \in S \cap T$$

Otherwise — when at least one factor is 0 — the product is 0. So

$$\boxed{1_S \cdot 1_T = 1_{S \cap T}.$$

1.4 Verification by Cases

Every $x \in D$ falls into exactly one of four regions; checking each:

Where x lies	$1_S(x)$	$1_T(x)$	$1_S(x) \cdot 1_T(x)$	$1_{S \cap T}(x)$
$S \cap T$	1	1	1	1
$S \setminus T$	1	0	0	0
$T \setminus S$	0	1	0	0
$D \setminus (S \cup T)$	0	0	0	0

Table 1.1

The fourth and fifth columns agree on every row, so the two functions are equal.

1.5 Remark — Boolean Operations as Pointwise Arithmetic

Characteristic functions translate set-theoretic operations into pointwise arithmetic on $\{0, 1\}$ -valued functions:

- **Intersection.** $1_{S \cap T} = 1_S \cdot 1_T$. (This problem.)
- **Union.** $1_{S \cup T} = 1_S + 1_T - 1_S \cdot 1_T$ (inclusion–exclusion).
- **Complement.** $1_{D \setminus S} = 1 - 1_S$.
- **Symmetric difference.** $1_{S \Delta T} = 1_S + 1_T - 2(1_S \cdot 1_T)$, which mod 2 is $1_S \oplus 1_T$ (XOR).

This is the bridge between set algebra and ordinary arithmetic. Logical “and” on $x \in S$ and $x \in T$ becomes multiplication of indicators; “or” becomes inclusion–exclusion; “not” becomes $1 - \cdot$. It is the same identification that makes indicator functions central to combinatorics, probability, and measure theory: every Boolean operation on subsets has a clean algebraic counterpart on indicators.